Good-Enough Vision at Birth: A Computational Demonstration of What Newborns Can See with Rod-Dominated, Low-Acuity Input

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Background

- Newborn infants have severely limited visual acuity and immature foveal cone systems, yet they show early sensitivity to face-like patterns and structured visual input. (Slater, 2002; Zele & Cao, 2015) This project asks whether such early competencies may be supported by coarse, rod-dominated visual information rather than adult-like high-resolution vision.

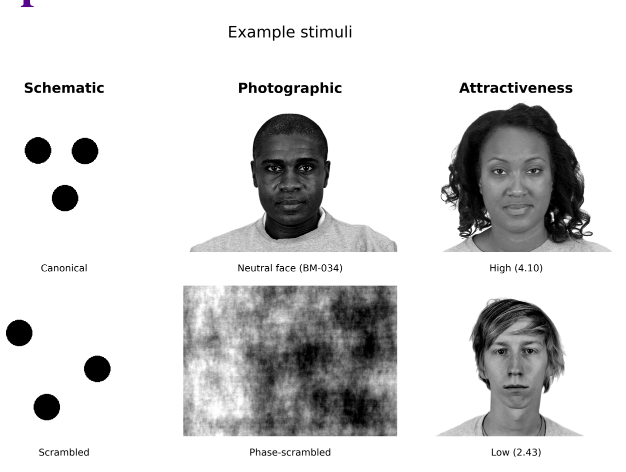
Research Question

- How much visual structure remains under rod-weighted, low-acuity conditions?
- Is this reduced visual information sufficient for face-relevant discrimination?

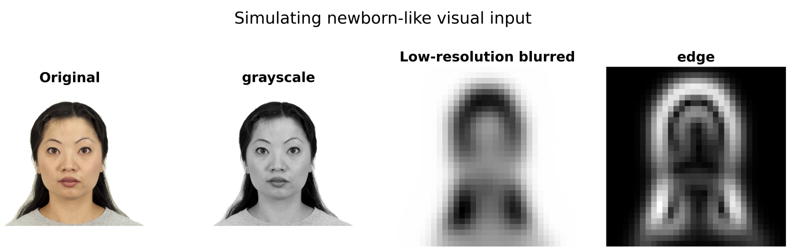
Methods

- Schematic canonical/scrambled faces and neutral faces from the Chicago Face Database (Ma et al., 2015)
- Newborn-like simulation: grayscale, normalization, 32×32 resize, Gaussian blur, optional Sobel edges
- Tasks: schematic vs. scrambled, face vs. phase-scrambled, attractiveness, cross-orientation generalization

Example Stimuli

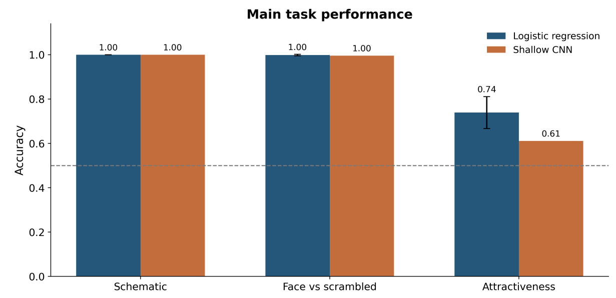


Modeling Pipeline



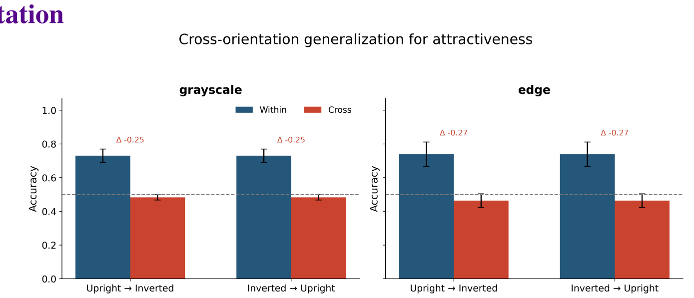
Percentile normalization (1.0 to 99.0), resize to 32x32, Gaussian blur σ=1.5, sobel edge extraction.

Results



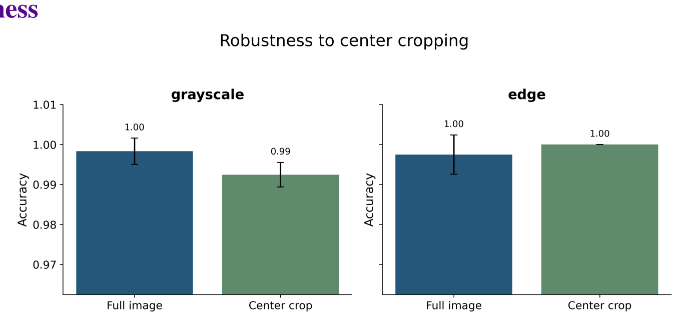
Best-performing linear input shown for each task

Cross-Orientation



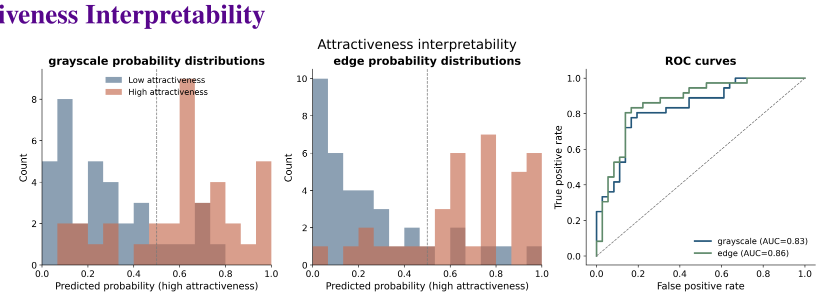
Within = within-orientation; Cross = cross-orientation.

Crop Robustness



Accuracy remains near ceiling across full image and center crop conditions.

Attractiveness Interpretability



Results Summary

- Canonical vs. scrambled schematic faces remained nearly perfectly discriminable after newborn-like degradation.
- Real faces vs. phase-scrambled controls also remained near ceiling, including under center cropping.
- Attractiveness classification remained above chance under degraded input, with the best linear model reaching $\sim$ **0.74** accuracy.
- Cross-orientation generalization collapsed ($\sim$ **0.73–0.74** within-orientation to $\sim$ **0.46–0.48** cross-orientation), indicating that the surviving attractiveness-related signal is strongly orientation-sensitive.

Discussion

- Coarse, newborn-like degraded input can still support multiple face-related discriminations.
- This reduces the need to assume adult-like, high-resolution vision to explain some early visual competencies.
- These models are used as proofs of informational sufficiency, not as literal mechanistic accounts of infant vision (Geirhos et al., 2019).

Limitations

- Simplified approximation of neonatal retinal input
- No explicit modeling of eye movements, attention, or learning
- The computational models are ideal-observer demonstrations, not biological process models

References

Geirhos, R., Rubisch, P., Michaelis, C., Bethge, M., Wichmann, F. A., & Brendel, W. (2019). ImageNet-trained CNNs are biased towards texture; increasing shape bias improves accuracy and robustness. In *International Conference on Learning Representations (ICLR)*.

Goren, C. C., Sarty, M., & Wu, P. Y. K. (1975). Visual following and pattern discrimination of face-like stimuli by newborn infants. *Pediatrics, 56*(4), 544–549.

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Website

GitHub